

Practical No. X: Stored Procedures in SQL Server

Aim: To create and execute Stored Procedures in SQL Server.

Theory

A Stored Procedure is a precompiled collection of SQL statements stored in the database and executed whenever required.

Question 1: Create a Stored Procedure to Display All Records from Products Table.

Create Table:

```
CREATE TABLE Products(ProductID INT PRIMARY KEY, ProductName VARCHAR(50), Price DECIMAL(10,2), Stock INT);
```

Insert Records:

```
INSERT INTO Products VALUES
```

```
(101,'Laptop',55000,10),(102,'Mobile',25000,20),(103,'Keyboard',1500,30);
```

Procedure:

```
CREATE PROCEDURE ShowProducts AS BEGIN SELECT * FROM Products; END;
```

Execute:

```
EXEC ShowProducts;
```

Question 2: Create a Stored Procedure with Input Parameter.

```
CREATE PROCEDURE GetProductByID @ProductID INT AS BEGIN SELECT * FROM Products WHERE ProductID=@ProductID; END;
```

Execute:

```
EXEC GetProductByID 101;
```

Question 3: Create a Stored Procedure to Insert Records.

```
CREATE PROCEDURE AddProduct (@ProductID INT,@ProductName VARCHAR(50),@Price DECIMAL(10,2),@Stock INT) AS BEGIN INSERT INTO Products VALUES(@ProductID,@ProductName,@Price,@Stock); END;
```

Execute:

```
EXEC AddProduct 104,'Mouse',800,25;
```

Question 4: Create a Stored Procedure to Update Product Price.

```
CREATE PROCEDURE UpdatePrice (@ProductID INT,@Price DECIMAL(10,2)) AS BEGIN UPDATE Products SET Price=@Price WHERE ProductID=@ProductID; END;
```

Execute:

```
EXEC UpdatePrice 101,60000;
```

Question 5: Create a Stored Procedure with Output Parameter.

```
CREATE PROCEDURE TotalProducts @Count INT OUTPUT AS BEGIN SELECT @Count=COUNT(*) FROM Products; END;
```

Execute:

```
DECLARE @Total INT; EXEC TotalProducts @Total OUTPUT; PRINT @Total;
```

Question 6: Create a Stored Procedure to Delete a Product.

```
CREATE PROCEDURE DeleteProduct @ProductID INT AS BEGIN DELETE FROM Products WHERE ProductID=@ProductID; END;
```

Execute:

```
EXEC DeleteProduct 104;
```

Viva Questions and Answers

Q1. What is a Stored Procedure?

Ans: A stored procedure is a collection of SQL statements stored in the database and executed as a single unit.

Q2. How do you execute a Stored Procedure?

Ans: EXEC ProcedureName;

Q3. What is an Input Parameter?

Ans: A value passed to a procedure during execution.

Q4. What is an Output Parameter?

Ans: A parameter used to return values from a procedure.

Q5. Difference between Procedure and Function?

Procedure can perform INSERT, UPDATE, DELETE operations whereas Function generally returns a value and is used in SQL expressions.

Result

Thus, various Stored Procedures using input parameters, output parameters, insert, update, delete, and retrieval operations were successfully created and executed.